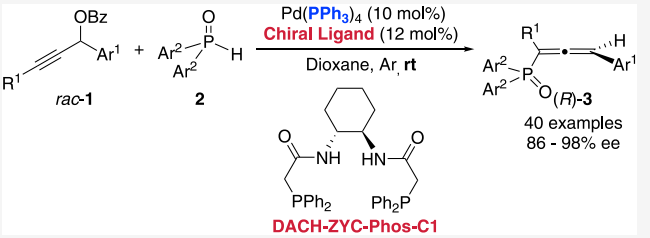
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Article

DACH-ZYC-Phos/Pd-Catalyzed Enantioselective Allenylation of Secondary Phosphine Oxides via Ligand Relay

Zhaoqiang Chen,‖ Huanan Wang,‖ Ping Du, Jiaao Zhao, Xue Zhang,* Hui Qian,* Junliang Zhang,* and Shengming Ma*

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ABSTRACT: Chiral allenyl phosphine oxides hold significant application value in organic synthesis, serving as efficient catalysts, ligands, and versatile synthons. However, the development of transition metal-catalyzed asymmetric synthesis for these compounds remains unexplored, primarily due to the facile tautomerization of phosphine oxides into trivalent R^1R^2P-OH species, which may severely deactivate transition metal catalysts through strong coordination interactions. Herein, a ligand relay strategy has been applied to address this challenge in the Pd-catalyzed enantioselective coupling reactions of propargylic benzoates and secondary phosphine oxides (SPOs) to afford chiral trisubstituted allenyl phosphine oxides with high ee. The ligand relay protocol involves the initial coordination of a palladium catalyst with readily available triphenylphosphine, followed by dynamic ligand exchange with the stretchable chiral ligand DACH-ZYC-Phos-C1 (**L26**), while PPh_3 does not work as a nonchiral ligand for palladium catalysts in this transformation. Chiral trivalent organophosphines with high ee have been obtained upon the reduction of allenyl phosphine oxides. Mechanistic studies revealed the nature of the kinetic resolution. DFT calculations provide a mechanistic rationale for the observed high enantioselectivity and superior catalytic efficiency, which is governed by the ligand **L26** through steric repulsions that selectively destabilize the disfavored oxidative addition transition states.



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INTRODUCTION

The introduction of a phosphine oxide group into molecules usually results in a dramatic change in their physical and chemical properties, which has attracted more and more attention. As a versatile functional unit, phosphine oxide groups have been widely applied in functional materials,[1] medicine,[2] catalysts,[3] and ligands for metal catalysts.[4] Of particular interest, phosphine oxides with an allene functionality are not only useful catalysts (Scheme 1a)[5] but also important synthons.[6−9] Recently, much attention has been paid to the synthesis and chemistry of allenes.[10−13] However, a transition metal-catalyzed asymmetric approach for allenyl phosphine oxides has not been developed due to the facile tautomerization of phosphine oxide into the trivalent R^1R^2P-OH, which may severely poison the transition metal via extensive coordination (Scheme 1b).[14] Specifically, the energy barrier for the isomerization of secondary phosphine oxides (SPOs) to trivalent R^1R^2P-OH is much lower than that of *H*-phosphonates.[15] Thus, the rate constant of SPO-to-R^1R^2P-OH is significantly higher than that of *H*-phosphonates (Scheme 1b), which reduces both the rate of formation and the catalytic activity of the envisioned chiral transition metal catalyst with the *M*-**L*** unit (Scheme 1c). Furthermore, such nonchiral metal-R^1R^2P-OH coordination may also promote nonchiral background reactions, thereby reducing enantioselectivity, which further makes the synthesis of chiral allenyl phosphine oxides more challenging.[16] To address this issue, we propose that a ligand relay strategy may provide a solution, whereby the transition metal catalyst first coordinates with an achiral ligand **L**, followed by ligand exchange with a chiral ligand **L***, to form the corresponding catalytically active species *M*-**L*** (Scheme 1c). To realize this proposed strategy, the exploration of readily available ligand **L** and a flexible chiral ligand **L*** emerges as a key strategy. Herein, we report such a ligand relay strategy for chiral palladium catalysis, enabling the catalytic enantioselective synthesis of chiral trisubstituted allenyl phosphine oxides through the coupling reactions of propargylic benzoates with SPOs. This is achieved via the synergistic cooperation of DACH-ZYC-Phos-C1 as the chiral entity and triphenylphosphine (PPh_3) as the relay ligand (Scheme 1d).

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Scheme 1. Importance of Phosphine Oxide Units and Enantioseletive Synthesis of Allenyl Phosphine Oxide

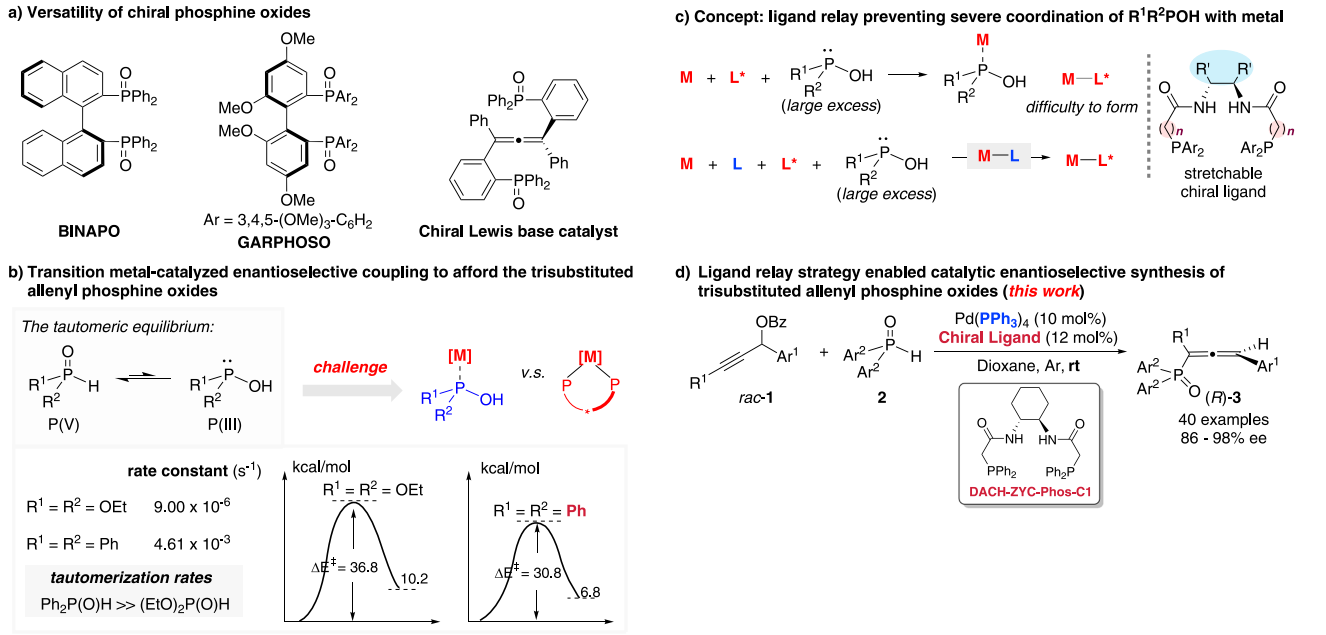
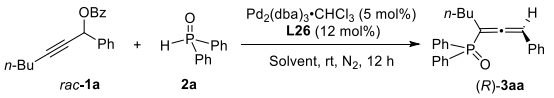


Table 1. Optimization of the Solvent[a]



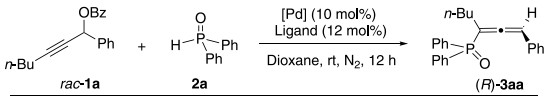
entry	Solvent	(*R*)-**3aa** (%) NMR yield	ee	recovered **1a**/**2a** (%)
1	THF	14	90	84/29
2	PhMe	0	-	97/0
3	MeCN	16	48	93/0
4	DCE	51	74	79/5
5	DCM	57	72	76/0
6	**Dioxane**	**33**	**92**	**86/39**
7	Anisole	16	93	92/0
8	DME	9	91	95/9

[a]The reactions were conducted with *rac*-**1a** (0.25 mmol), **2a** (0.1 mmol), $Pd_2(dba)_3$·$CHCl_3$ (5 mol %), and **L26** (12 mol %) in solvent (1.0 mL) at room temperature under N_2 atmosphere. The yield of (*R*)-**3aa** was determined by ^{1}H NMR analysis and calculated based on the limiting reagent **2a**. The recoveries of **1a** and **2a** were determined by ^{1}H NMR analysis and calculated based on their own usage. The ee value was determined by HPLC analysis.

RESULTS AND DISCUSSION

Control of Enantioselectivity. We commenced our research on the coupling reaction of 1-phenylhept-2-ynyl benzoate **1a** and diphenylphosphine oxide **2a** with $Pd_2(dba)_3$·$CHCl_3$ as the precatalyst to examine the effect of chiral phosphine ligands in THF (Scheme 2) by applying the reaction conditions we recently developed for the asymmetric coupling reaction between propargylic derivatives and *H*-phosphonates.[17] As expected, such a protocol was inapplicable for secondary phosphine oxide (SPO): no allenyl phosphine oxide product was obtained when diphenylphosphine oxide was used instead of *H*-phosphonates with the well-recognized, commercially available Trost ligand **L1** (Scheme 2). The reason is quite simple: as already discussed, the tautomeriza-

Table 2. Optimization of the Palladium Source[a,b,c,d]

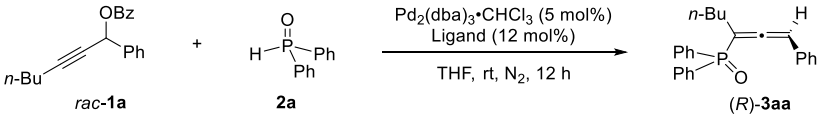


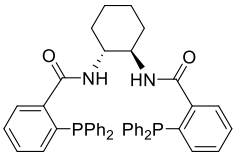
entry	[Pd]	Ligand	(*R*)-**3aa** (%) NMR yield	ee	recovered **1a**/**2a** (%)
1	$Pd(OAc)_2$	**L26**	0	-	95 / 46
2	$PdCl_2$	**L26**	0	-	96 / 61
3	$Pd(acac)_2$	**L26**	0	-	91 / 70
4	$Pd(PPh_3)_2Cl_2$	**L26**	0	-	96 / 62
5	$Pd_2(dba)_3$	**L26**	29	94	83 / 42
6	$Pd(dmdba)_2$	**L26**	47	94	81 / 28
7	$Pd(dba)_2$	**L26**	56	93	77 / 19
8	$Pd_2(dba)_3$	**L26** + PPh_3 (40 mol%)	40	93	78 / 29
9	$Pd(dmdba)_2$	**L26** + PPh_3 (40 mol%)	69	92	70 / 16
10	$Pd(dba)_2$	**L26** + PPh_3 (40 mol%)	83	90	62 / 0
11	$Pd(dba)_2$	**L26** + PPh_3 (30 mol%)	77	91	69 / 6
12	$Pd(dba)_2$	**L26** + PPh_3 (10 mol%)	77	91	69 / 3
13[b]	$Pd(dba)_2$	**L26** + **L-1** (40 mol%)	59	90	73 / 13
14[c]	$Pd(dba)_2$	**L26** + **L-2** (40 mol%)	55	90	77 / 25
15[d]	$Pd(dba)_2$	**L26** + **L-3** (40 mol%)	55	93	77 / 25
16	$Pd(PPh_3)_4$	**L26**	88	90	64 / 0
17	$Pd(PPh_3)_4$	**L1**	2	-	96 / 75
18	$Pd(PPh_3)_4$	-	-	-	95 / 42

[a]The reactions were conducted with *rac*-**1a** (0.25 mmol), **2a** (0.1 mmol), [Pd] (10 mol %), and **L26** (12 mol %) in dioxane (2.0 mL) at room temperature under N_2 atmosphere. The yield of (*R*)-**3aa** was determined by ^{1}H NMR analysis and calculated based on the limiting reagent **2a**. The recoveries of **1a** and **2a** were determined by ^{1}H NMR analysis and calculated based on their own usage. The ee value was determined by HPLC analysis. [b]**L-1** = tris(4-methylphenyl)-phosphine. [c]**L-2** = tris(4-methoxyphenyl)phosphine. [d]**L-3** = tris(4-(trifluoromethyl)phenyl)phosphine.

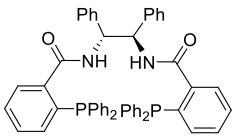
https://doi.org/10.1021/jacs.5c07465
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Scheme 2. Ligand Screening[a]

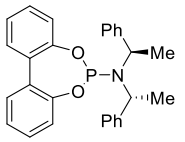




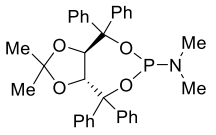
L1
no desired pdt
91% recovery of **1a**
39% recovery of **2a**



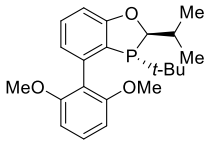
L2
no desired pdt
93% recovery of **1a**
63% recovery of **2a**



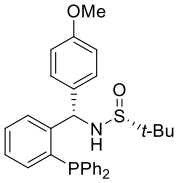
L3
no desired pdt
92% recovery of **1a**
63% recovery of **2a**



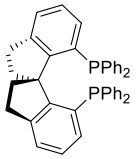
L4
no desired pdt
97% recovery of **1a**
64% recovery of **2a**



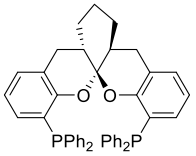
L5
no desired pdt
90% recovery of **1a**
53% recovery of **2a**



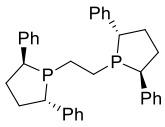
L6
no desired pdt
97% recovery of **1a**
40% recovery of **2a**



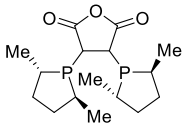
L7
no desired pdt
95% recovery of **1a**
37% recovery of **2a**



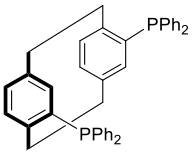
L8
no desired pdt
96% recovery of **1a**
37% recovery of **2a**



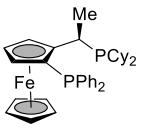
L9
no desired pdt
95% recovery of **1a**
74% recovery of **2a**



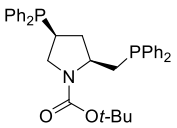
L10
no desired pdt
94% recovery of **1a**
61% recovery of **2a**



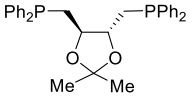
L11
15%, 33% ee
87% recovery of **1a**
47% recovery of **2a**



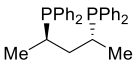
L12
18%, 2% ee
88% recovery of **1a**
50% recovery of **2a**



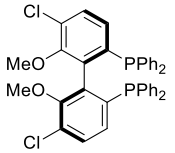
L13
16%, 14% ee
87% recovery of **1a**
56% recovery of **2a**



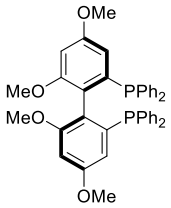
L14
81%, 6% ee
51% recovery of **1a**



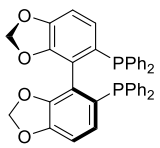
L15
72%, 10% ee
55% recovery of **1a**



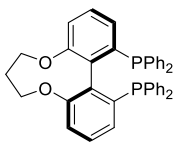
L16
81%, 3% ee
60% recovery of **1a**



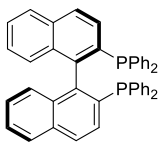
L17
32%, 19% ee
75% recovery of **1a**
37% recovery of **2a**



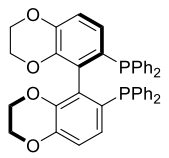
L18
58%, -26% ee
66% recovery of **1a**
17% recovery of **2a**



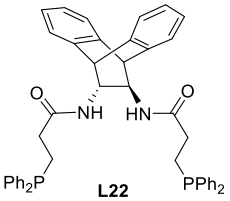
L19
37%, 30% ee
83% recovery of **1a**
45% recovery of **2a**



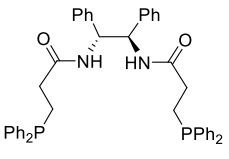
L20
24%, 46% ee
85% recovery of **1a**
41% recovery of **2a**



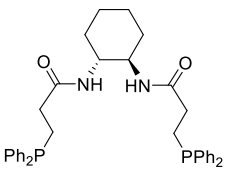
L21
34%, 46% ee
84% recovery of **1a**
38% recovery of **2a**



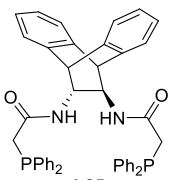
L22
trace
97% recovery of **1a**
65% recovery of **2a**



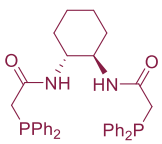
L23
9%, 6% ee
90% recovery of **1a**
54% recovery of **2a**



L24
6%, 1% ee
93% recovery of **1a**
55% recovery of **2a**



L25
no desired pdt
90% recovery of **1a**
30% recovery of **2a**



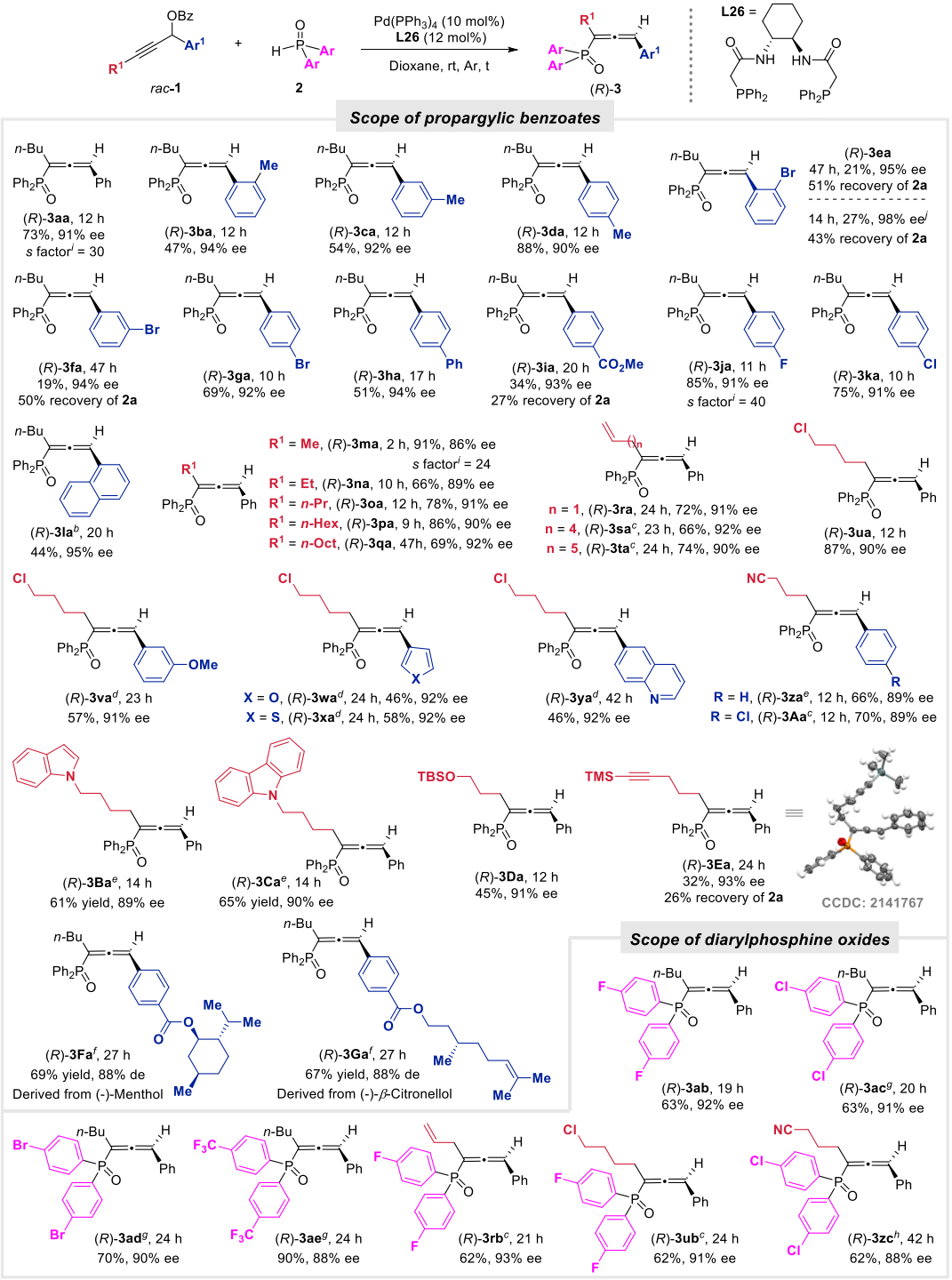
L26
14%, 90% ee
84% recovery of **1a**
32% recovery of **2a**

[a]Reaction conditions: *rac*-**1a** (0.25 mmol), **2a** (0.1 mmol), Pd(dba)·CHCl (5 mol %), and ligand (12 mol %) in THF (1.0 mL) at room temperature under N atmosphere. The yield of (*R*)-**3aa** was determined by ^{1}H NMR analysis and calculated based on the limiting reagent **2a**. The ee value was determined by HPLC analysis.

tion of diphenylphosphine oxide has a much lower energy barrier to form the trivalent Ph_2P-OH compared to *H*-phosphonates as shown in Scheme 1b, thus poisoning the catalytic activity of palladium via coordination and resulting in a higher recovery of **1a**. Then, another Trost ligand **L2** was tested, but still no desired product (*R*)-**3aa** could be obtained. The use of monophosphine ligands, such as phosphoramidite ligands **L3** and **L4**, BIDIME **L5**, and Sadphos **L6**, was also inefficient. We also examined various bis-phosphine ligands: **L7**-**L8** with spiro chirality and **L9**−**L10** with central chirality; however, the reaction did not yield any allenylic product. The use of **L11** and **L12** with planar chirality and **L13** with central chirality also led to low yields. High yields were obtained when DIOP (**L14**) was applied as the ligand; however, the

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Scheme 3. Substrate Scope[abcdefghij]

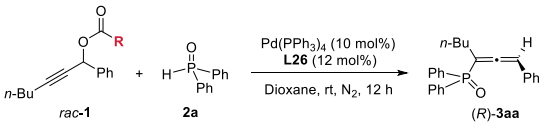


[a]Reaction conditions: *rac*-**1** (2.5 mmol), **2** (1.0 mmol), $Pd(PPh_3)_4$ (10 mol %), and **L26** (12 mol %) in dioxane (30 mL) at room temperature under an Ar atmosphere. The yields presented are isolated yields, calculated based on the limiting reagent **2**. The recoveries of **2** were determined by ^{1}H NMR analysis and calculated based on their own usage. The ee value was determined by HPLC analysis. [b]With $Pd(PPh_3)_4$ (16 mol %), and **L26** (20 mol %), 0.9 mmol of **2a**. [c]With *rac*-**1** (2.6 mmol), 25 °C. [d]With *rac*-**1** (2.6 mmol), $Pd_2(dba)_3.CHCl_3$ (5 mol %), **L26** (12 mol %), 25 °C. [e]25 °C. [f]With *rac*-**1** (0.52 mmol), **2** (0.2 mmol), $Pd(PPh_3)_4$ (15 mol %), **L26** (18 mol %), 25 °C. [g]With $Pd(PPh_3)_4$ (15 mol %), **L26** (18 mol %). [h]With *rac*-**1** (2.6 mmol), $Pd_2(dba)_3{\cdot}CHCl_3$ (7.5 mol %), **L26** (18 mol %), 25 °C. [i]The reaction was conducted with *rac*-**1** (0.2 mmol) and **2** (0.1 mmol). *s* factor = $\ln[(1 - ee_{sub.})/(1 + ee_{sub.}/ee_{pdt.})]/\ln[(1 + ee_{sub.})/(1 + ee_{sub.}/ee_{pdt.})]$. [j]3.5 mmol of *rac*-**1e** was applied.

enantioselectivity was very low. BDPP (**L15**) also failed to control the enantioselectivity. When biarylphosphine ligands **L16**–**L21** were tested, the desired product (*R*)-**3aa** was obtained in 24–81% yields with poor enantioselectivities. We then turned our attention to a new type of ligand with a flexible backbone[18] and a fixed two-carbon chain linker (*n* = 2, **L22**–**L24**), but still with no satisfactory results. Given the stretchable nature of the ligands, we tuned the carbon chain

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Table 3. Optimization of the Leaving Group[a]



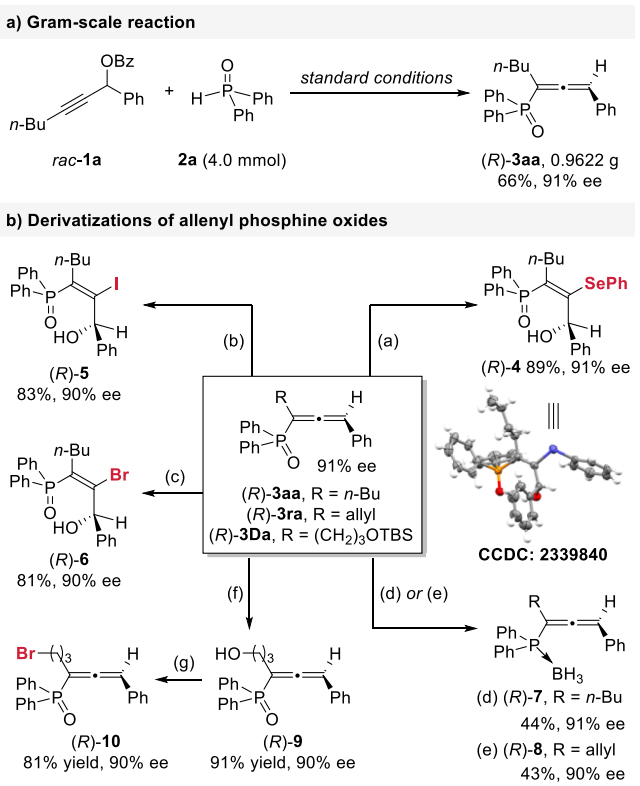
entry	R	(R)-3aa (%) NMR yield	(R)-3aa (%) ee	recovered 1a/2a (%)
1	Ph	88	90	64 / 0
2	Me	50	88	78 / 0
3	Adamantyl	26	96	78 / 3
4	2-Naphthyl	58	92	60 / 12
5	OMe	74	79	59 / 0
6	OBn	79	75	58 / 0

[a]The reactions were conducted with *rac*-**1** (0.25 mmol), **2a** (0.1 mmol), $Pd(PPh_3)_4$ (10 mol %), and **L26** (12 mol %) in dioxane (2.0 mL) at room temperature under N_2 atmosphere. The yield of (*R*)-**3aa** was determined by ^1H NMR analysis and calculated based on the limiting reagent **2a**. The recoveries of **1a** and **2a** were determined by ^1H NMR analysis and calculated based on their own usage. The ee value was determined by HPLC analysis.

to adjust the size of the chiral pocket, and the DACH-ZYC-Phos ligand with a one-carbon linker, **L26,** was observed to control the enantioselectivity highly efficiently, achieving 90% ee; however, the yield was very low at 14%. The solvent effect was then examined with **L26** as the ligand (Table 1). Toluene failed to form the target product (Table 1, entry 2). In MeCN, DCE, or DCM, the reaction afforded the product with lower enantioselectivity (Table 1, entries 3−5). Further solvent screening led to the observation of dioxane as the optimal choice, affording the product (*R*)-**3aa** in 92% ee, albeit still with a mere 33% yield (Table 1, entry 6). Thus, both the concentration and reactivity of the chiral palladium catalyst are key here.

The Ligand Relay Strategy. Subsequent palladium source screening in dioxane (Table 2) indicated that none of the tested Pd(II) precatalysts could catalyze this coupling reaction (Table 2, entries 1−4). The Pd(0) catalysts, such as $Pd_2(dba)_3$, $Pd(dba)_2$, and $Pd(dmdba)_2$, afforded (*R*)-**3aa** with high enantioselectivities but unsatisfactory yields. Then, the readily available triphenylphosphine was introduced as a relay ligand. For the reaction with $Pd_2(dba)_3$ as the precatalyst, the yield of (*R*)-**3aa** improved to 40% with similar enantioselectivity when 40 mol % of PPh_3 was added (Table 2, entry 8 *vs* entry 5). A similar effect was observed for the reaction with $Pd(dmdba)_2$ as the precatalyst (Table 2, entry 9 *vs* entry 6). To our delight, the yield of (*R*)-**3aa** was improved to 83% with 90% ee for the combination of $Pd(dba)_2$ with PPh_3 and **L26** (Table 2, entry 10). When the loading of PPh_3 was reduced to 30% or 10%, the yields of (*R*)-**3aa** slightly dropped to 77% with similar enantioselectivity (Table 2, entries 11 and 12). Other monophosphines, such as tris(4-methylphenyl)phosphine **L-1**, tris(4-methoxyphenyl)phosphine **L-2**, and tris(4-(trifluoromethyl)phenyl)phosphine **L-3,** were also examined (Table 2, entries 13−15). Both the yield and enantioselectivity of (*R*)-**3aa** were similar to those of the reaction in the absence of PPh_3 (Table 2, entry 7). The most commonly available PPh_3 proved to be the optimal relay ligand. $Pd(PPh_3)_4$, in which the palladium atom was precoordinated with PPh_3, was also tested, and (*R*)-**3aa** was finally obtained in 88% yield with 90% ee (Table 2, entry 16). These results indicated the significance of

Scheme 4. Gram-scale Reaction and Synthetic Applications[abcdefg]



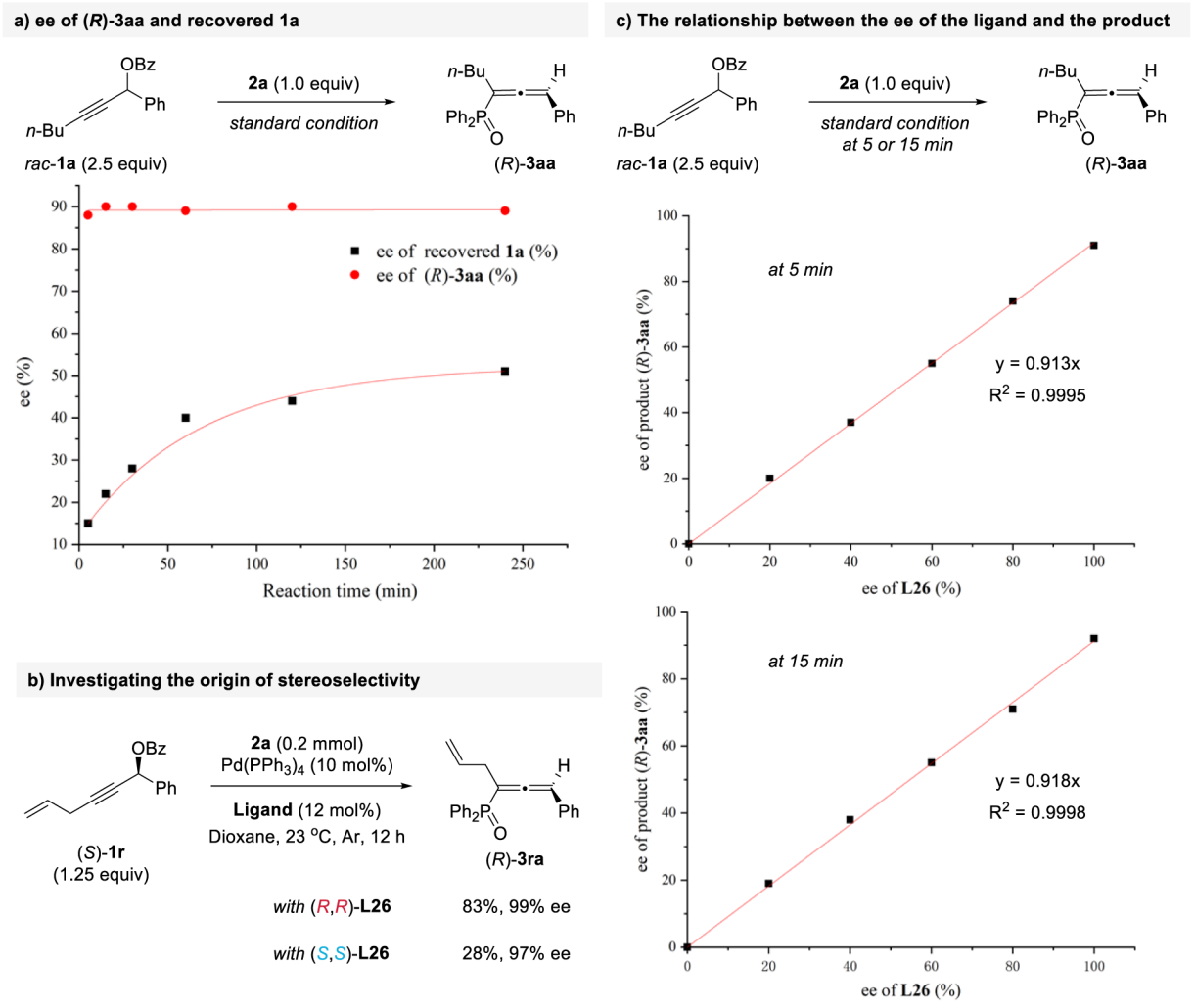
[a]Reaction conditions for synthetic applications: (*R*)-**3aa** (0.2 mmol), PhSeCl (0.3 mmol), $MeCN/H_2O$ = 4.0/0.2 mL, rt, 70 min. [b](*R*)-**3aa** (0.2 mmol), I_2 (0.8 mmol), and $MeCN/H_2O$ = 2.0/5.0 mL, rt, 16 h. [c](*R*)-**3aa** (0.2 mmol), $CuBr_2$ (0.6 mmol), TBHP (0.4 mmol), and THF (1.0 mL), 0 °C, 17 h. [d](*R*)-**3aa** (0.2 mmol), MeOTf (0.4 mmol), $LiAlH_4$ (0.325 mmol), BH_3·THF (0.4 mmol), DME (3.0 mL), −70 to 0 °C, 10.5 h. [e](*R*)-**3ra** (0.2 mmol), MeOTf (0.4 mmol), $LiAlH_4$ (0.325 mmol), BH_3·THF (0.4 mmol), DME (3.0 mL), −70 to 0 °C, 10 h. [f](*R*)-**3Da** (0.8 mmol), HFPy (2.3 equiv), MeCN, rt, 1.5 h. [g](*R*)-**9** (0.2 mmol), CBr_4 (1.3 equiv), PPh_3 (1.5 equiv), DCM, 0 °C, 1.5 h.

PPh_3 in preventing palladium from being poisoned. We next conducted two control experiments (Table 2, entries 17 and 18) and found that essentially no reaction was observed when **L1** was employed or in the absence of **L26**. These findings further underscore the superiority of ZYC-Phos **L26** and the importance of PPh_3. Subsequent optimization of the leaving group led to the conclusion that benzoate was still the best (Table 3).

Substrate Scope. With the optimized reaction conditions in hand, we commenced exploring the substrate scope (Scheme 3). A variety of propargylic benzoates bearing different Ar^1 with electron-donating and electron-withdrawing groups could be employed. It is noteworthy that the steric effect plays an important role, as the propargylic benzoates with a Me or Br substituent at the *p*-position of aryl (Ar^1) possessed better reactivity than the *o*- and *m*-positions to afford the corresponding products (*R*)-**3ba**−(*R*)-**3ga**. In addition, an electronic effect of the *para*-substituents on the Ar^1 ring was observed: the reactivity was higher with electron-donating groups (**1d, 1g, 1h,** and **1i**). Other halogen substituents (F, Cl) were also tolerated, forming the trisubstituted allenylic products (*R*)-**3ja**−(*R*)-**3ka**. Replacing the Ar^1 group with

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Scheme 5. Mechanistic Studies



naphthyl made no difference, furnishing the product (*R*)-**3la** in moderate yield with excellent ee. The reaction also enjoys an excellent generality for alkyl $\mathbf{R^1}$ chains of varying lengths, providing the target products (*R*)-**3ma**−(*R*)-**3qa** in good yields, with better enantioselectivity achieved as the alkyl chain length increased. Meanwhile, a wide range of synthetically versatile functional groups, such as terminal alkenes with varying carbon chain lengths (**1r**−**1t**), halogen (**1u**−**1y**), furan (**1w**), thiene (**1x**), quinoline (**1y**), cyano (**1z**−**1A**), indole (**1B**), carbazole (**1C**), silyl-protected alcohol (**1D**), and alkynyl (**1E**), were accommodated, affording the corresponding products (*R*)-**3ra**−(*R*)-**3Ea** with yields of 32−87%. The low yield of (*R*)-**3Ea** might be attributed to the coordination of the carbon−carbon triple bond to palladium, leading to a reduced reactivity of the substrate. Additionally, the current protocol was also applicable for highly selective late-stage enantioselective C−P coupling of bioactive complexes, such as menthol and cholesterol, supplying the corresponding allene phosphine oxides (*R*)-**3Fa** and (*R*)-**3Ga** in good yields and excellent ees. The absolute configuration of all the products was confirmed through single-crystal X-ray diffraction analysis of (*R*)-**3Ea**. Furthermore, SPOs bearing different substituents, such as halides (**2b**−**2d**) and trifluoromethyl (**2e**) all performed well, affording the allenyl phosphine oxides (*R*)-**3ab**−(*R*)-**3ae**, (*R*)-**3rb**, (*R*)-**3ub**, and (*R*)-**3zc** in decent yields.

Synthetic Applications. In order to demonstrate the synthetic potential of the current strategy, a gram-scale reaction was carried out, affording the product (*R*)-**3aa** in 66% yield with 91% ee (Scheme 4a). Electrophilic PhSeCl reacted with (*R*)-**3aa** in a mixed solvent of MeCN and water to yield (*R*)-**4** without the loss of ee. A single-crystal X-ray diffraction analysis was performed to ascertain the absolute configuration of (*R*)-**4**, which further confirmed the *R*-configuration in the allenyl products based on a previous report.[7b] Halohydroxylation of the electron-rich C═C bond afforded the multisubstituted olefins (*R*)-**5** or (*R*)-**6** through highly regioselective addition with allene.[7a,7e] Due to the introduction of a halogen atom, (*R*)-**5** and (*R*)-**6** can be further converted into diverse, versatile products. We also realized the reduction of allenyl phosphine oxide (*R*)-**3aa** as well as (*R*)-**3ra** to afford complexes of trivalent phosphorus products with borane, (*R*)-**7** and (*R*)-**8** in excellent ee.[19] Treatment of (*R*)-**3Da** with HFPy at room temperature resulted in deprotection affording allenol (*R*)-**9** in 91% yield with 90% ee. Subsequently, the Appel reaction of (*R*)-**9** produced allenyl bromide (*R*)-**10** with a yield of 81% and ee of 90% (Scheme 4b).

To gain insight into the mechanism, the ee values of (*R*)-**3aa** and recovered **1a** were monitored, which unveiled a kinetic resolution process (Scheme 5a). We also carried out the

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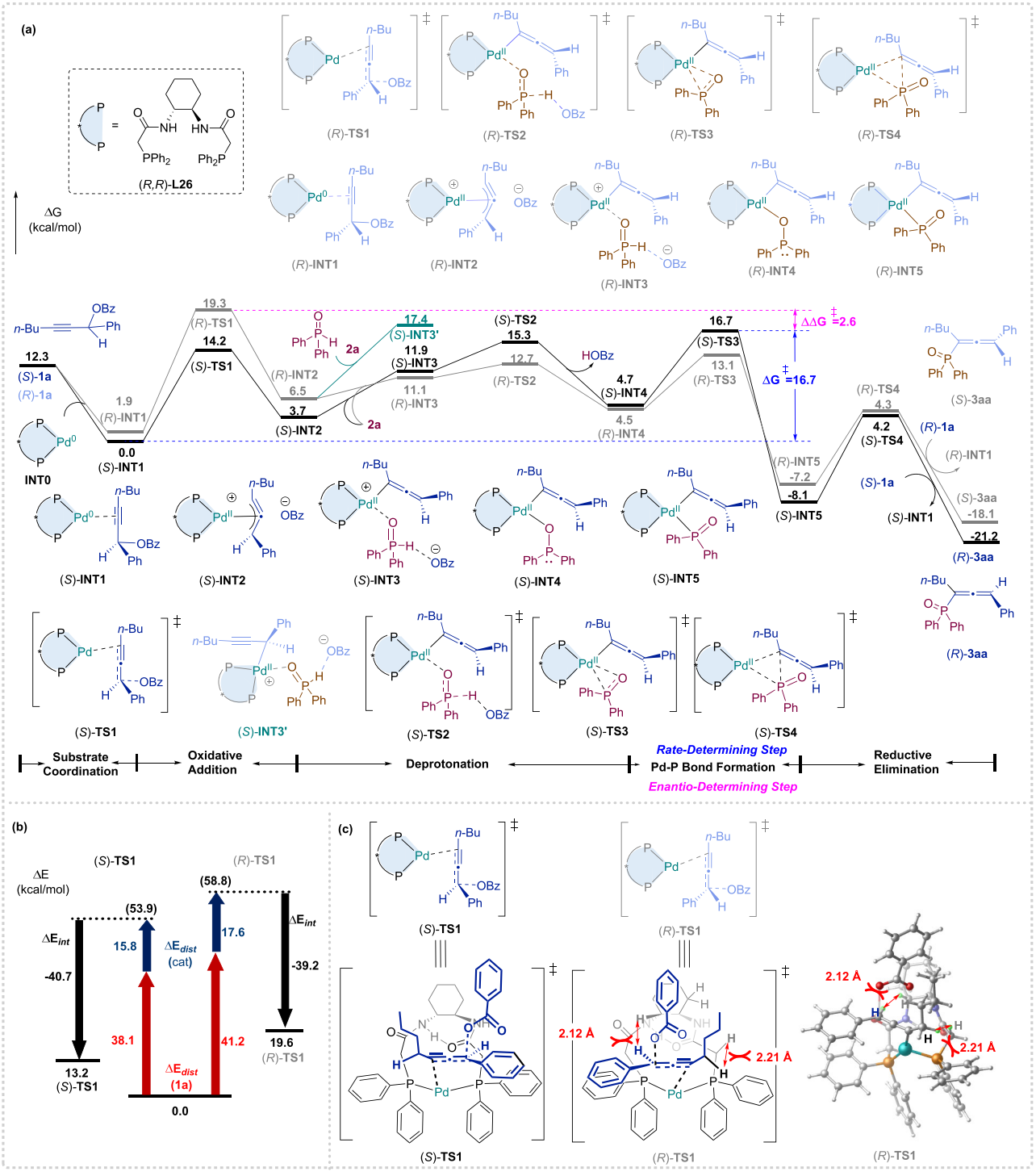


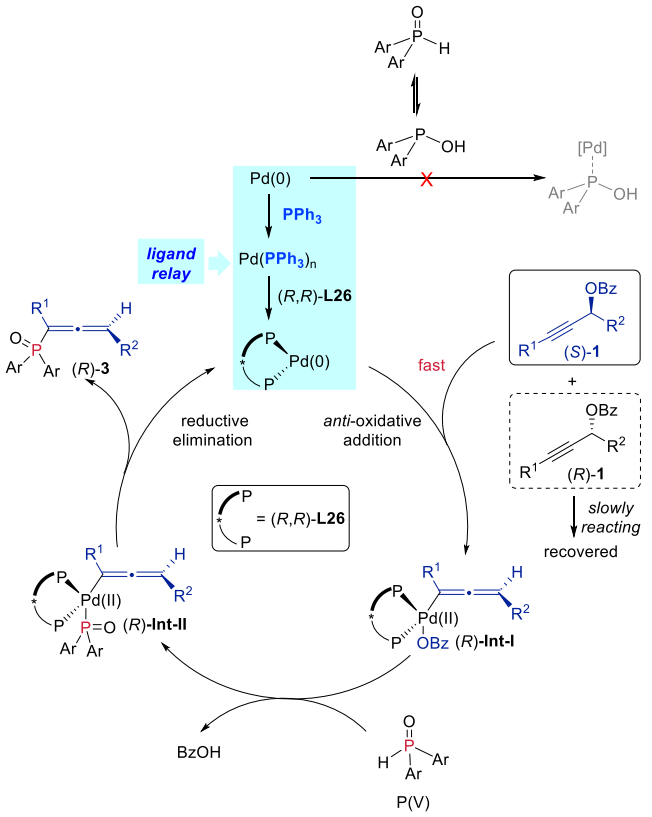
Figure 1. DFT calculations. (a) Free energy profiles of Pd-catalyzed enantioselective coupling reaction of **1a** and **2a**, using (*R*,*R*)-**L26** as the ligand. (b) Distortion−Interaction Analysis of (*S*)-**TS1** and (*R*)-**TS1**. (c) Steric effect analysis of (*S*)-**TS1** and (*R*)-**TS1**.

reactions using enantiopure (*S*)-**1r** with both enantiomers of **L26** (Scheme 5b): 83% of (*R*)-**3ra** was obtained with 99% ee when (*R*,*R*)-**L26** was applied, and only 28% of the same enantiomer (*R*)-**3ra** was obtained when (*S*,*S*)-**L26** was applied for the same reaction time of 12 h. These results indicated that (*R*,*R*)-**L26** matched with (*S*)-**1r**, further confirming the nature of the kinetic resolution process. In addition, the relationship between the ee of the ligand and the product was studied at 5 and 15 min (Scheme 5c), demonstrating a linear effect, which indicated that one ligand coordinated with one palladium atom.

Furthermore, DFT calculations were performed to elucidate the origin of enantioselectivity control by probing the reaction between **1a** and **2a**, employing (*R*,*R*)-**L26** as the ligand (see the Supporting Information for computational details). As depicted in Figure 1a, the Pd(0)[(*R*,*R*)-**L26**] species **INT0**, derived from the ligand exchange of the precatalyst, serves as the active catalyst, which is capable of activating (*S*)-**1a** through coordination to its triple bond to provide (*S*)-**INT1**.

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Scheme 6. Proposed Mechanism



The subsequent *anti*-oxidative addition of (*S*)-**INT1** proceeds via (*S*)-**TS1**, which requires an activation energy of 14.2 kcal/mol, resulting in the formation of the π-propargyl-Pd intermediate (*S*)-**INT2**. Further coordination of diphenylphosphine oxide **2a** to the Pd(II) center leads to the generation of either the η-1 coordinated allenyl palladium intermediate (*S*)-**INT3** or the propargyl palladium intermediate (*S*)-**INT3′**. The latter, (*S*)- **INT3′**, is thermodynamically less stable than (*S*)-**INT3** by 5.5 kcal/mol and thus energetically unfavorable. Deprotonation of (*S*)-**INT3** by the OBz^- anion then proceeds through (*S*)-**TS2**, with an activation barrier of 15.3 kcal/mol, yielding (*S*)-**INT4** and simultaneously facilitating the transformation of pentavalent to trivalent phosphine. Subsequently, ligand exchange between phosphine and oxygen on the palladium center in (*S*)-**INT4** proceeds via (*S*)-**TS3**, requiring a free energy of 16.7 kcal/mol. This step produces the phosphoryl palladium intermediate (*S*)-**INT5**, which is exergonic by 8.1 kcal/mol (relative to (*S*)-**INT1**). Finally, the reductive elimination of (*S*)-**INT5** occurs with an energy barrier of 12.3 kcal/mol ((*S*)-**TS4**), generating the *R*-configured major product, allenyl phosphine oxide (*R*)-**3aa**, while simultaneously introducing another molecule of (*S*)-**1a** to regenerate (*S*)-**INT1**, thereby completing the catalytic cycle. Consequently, the Pd−P bond formation step via (*S*)-**TS3** is identified as both the rate-limiting and enantiodetermining step, with the highest free energy barrier of 16.7 kcal/mol throughout the whole process. In contrast, the free energy profile corresponding to the minor enantiomer of (*S*)-**3aa** (gray line in Figure 1a), yielded from the mismatched combination of (*S*)-**1a** with (*R,R*)-**L26**, exhibits a similar pathway but with higher energy demands, particularly in the oxidative addition step ((*R*)-**TS1**, 19.3 kcal/mol). This indicates that oxidative addition is both the rate-limiting and enantioselectivity-determining step for the (*S*)-**3aa** formation. The calculated enantioselectivity, based on the free energy difference of 2.6 kcal/mol between (*R*)-**TS1** and (*S*)-**TS3**, predicts a 98% enantiomeric excess (ee) favoring the *R* enantiomer, which aligns well with the experimental ee value of 91%.

Further distortion-interaction energy analysis of the transition states (*S*)-**TS1** and (*R*)-**TS1** was conducted to gain in-depth knowledge of the enantioselectivity control (Figure 1b). The computed interaction energy (ΔE_{int}) between fragments **INT0** and **1a** is similar for both (*S*)-**TS1** (ΔE_{int} = −40.7 kcal/mol) and (*R*)-**TS1** (ΔE_{int} = −39.2 kcal/mol). In contrast, a significant difference is observed for the distortion energy (ΔE_{dist}) between the (*R,R*)-**L26** ligand and (*R*)- or (*S*)-**1a** required to adopt the geometry in the transition states. Specifically, (*S*)-**TS1** exhibits a distortion energy of 53.9 kcal/mol, while (*R*)-**TS1** requires a higher distortion energy of 58.8 kcal/mol. This difference is primarily attributed to the greater energetic penalty imposed on the **1a** fragment in (*R*)-**TS1** ($\Delta E_{dist(1a)}$ = 38.1 kcal/mol for (*S*)-**TS1**; $\Delta E_{dist(1a)}$ = 41.2 kcal/mol for (*R*)-**TS1**), which arises from steric clashes between the (*R,R*)-**L26** ligand and the substrate (*R*)-**1a**. As illustrated in Figure 1c, the steric repulsions between the H atoms of (*R,R*)-**L26** and (*R*)-**1a** (with distances of 2.12 and 2.21 Å) in (*R*)-**TS1** destabilize the transition state. In the absence of the steric hindrance, (*S*)-**TS1** is more favorable than (*R*)-**TS1** by 6.4 kcal/mol in overall electronic energy, highlighting the critical role of the steric effect of the ligand with the substrate in governing the enantioselectivity of the reaction.

According to these results, a possible reaction pathway was proposed, as shown in Scheme 6. The Pd(0) catalyst first coordinated with PPh_3 followed by ligand exchange with the chiral ligand **L26** to form the active catalyst Pd-**L26**, which would undergo a much faster *anti*-oxidative addition[20] with (*S*)-**1** to give the allenyl Pd(II) intermediate (*R*)-**Int-I**. Subsequent ligand exchange between (*R*)-**Int-I** and SPO would form (*R*)-**Int-II** while releasing one molecule of benzoic acid. After reductive elimination, the desired product (*R*)-**3** was produced along with the chiral Pd(0) catalyst.

CONCLUSIONS

In summary, we have successfully developed a relay ligand effect between PPh_3 and DACH-ZYC-Phos, enabling an efficient Pd-catalyzed asymmetric coupling of propargylic benzoates and secondary phosphine oxides (SPOs) for the synthesis of chiral allenyl phosphine oxides with excellent enantioselectivity. The corresponding products could be further converted into various phosphorus compounds without racemization. Importantly, the chiral allenyl phosphine oxide could be reduced to the chiral allenyl phosphine, which may have potential applications in asymmetric catalysis. Preliminary mechanistic studies indicated the kinetic resolution nature of this process. The observed enantioselectivity has been rationalized by DFT studies. Further studies on the mechanism and related reactions are currently ongoing in our laboratory.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.5c07465.

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Detailed experimental procedures, characterization of new compounds (PDF) Crystallographic information file for (*R*)-**3Ea** (CIF) Crystallographic information file for (*R*)-**4** (PDF)

Accession Codes

Deposition numbers 2141767 and 2339840 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Center (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Author Contributions

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Notes

The authors declare no competing financial interest.

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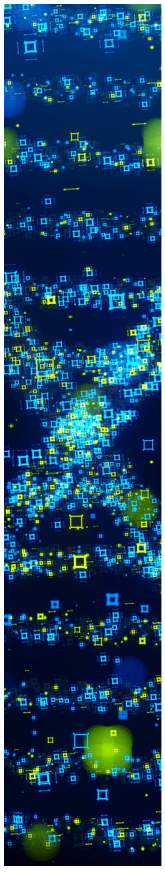
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